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Paleomagnetism and geochronology of the Bebedouro cap carbonate: evidence for continental-scale Cambrian remagnetization in the São Francisco craton, Brazil

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Abstract

New Pb–Pb ages, rock magnetic, and paleomagnetic data were obtained on Neoproterozoic carbonates of the Salitre Formation which conformably overlie the glacial rocks of the Bebedouro Formation in undeformed areas of the Irecê basin, São Francisco craton. These data give new supporting evidence for a large-scale remagnetization and resetting of the U–Pb isotopic system at Cambrian times along the São Francisco and Irecê basins, which cover more than 300,000 km^2 of the São Francisco craton surface. A ^{207}Pb–^{206}Pb isochron age of 514 ± 33 Ma was on undeformed, stromatolitic carbonates from the Salitre Formation. Although samples from other sites give almost non-radiogenic Pb signatures, they fall at the lower end of this line. Rock magnetic properties obtained for most of the sites are typical of remagnetized carbonates, and include: wasp-waisted hysteresis loops, contradictory Lowrie–Fuller and Cisowski tests, and anomalously high hysteresis ratios. Some samples, however, have presented only some of these magnetic characteristics. Four magnetic components (A, B, C, and D) were isolated from the Salitre carbonates. Three of them (A, B, and C) are comparable to previous results from the correlative Bambuí carbonates, which crop out ca. 600 km southward. The component C, carried by fine-grained SD-PSD magnetite, was identified at 23 sites. It directs steeply positive to the northeast, giving a paleomagnetic pole (SaC) at 33°N, 323°E ($\alpha_{95} = 4.0°$, $k = 23.6$). The SaC pole overlaps with that obtained from component C of Bambuí carbonates (BaC pole: 30°N, 321°E, $\alpha_{95} = 3.8°$). The paleopoles from component B of Salitre (SaB) and Bambuí (BaB) also lie close but are statistically different. After rotation of South America to Africa, all these poles fall close to the ∼520 Ma segment of the Gondwana apparent polar wander path, suggesting that the resetting of the isotopic and magnetic systems in the Neoproterozoic carbonates of the São Francisco craton occurred at that time. The single polarity directions found along the whole sedimentary succession in both areas give additional support for rapid remagnetization events. We suggest that paleomagnetic data and most of the Pb–Pb isochron ages in the São Francisco carbonates recorded a

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regional-scale fluid migration event, in the aftermath of the Brasiliano (Panafrican) collision. This fluid-flow event may also have concurred for the genesis of the Pb–Zn mineralizations found across the basin.
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1. Introduction

The carbonate rocks that typically overlie glacial sediments in Neoproterozoic successions are potential candidates for paleomagnetic studies (Park, 1997; Li, 2000). Poles obtained from these rocks would be useful to complete the Neoproterozoic paleomagnetic database which is still quite poor in most continents (Meert and Powell, 2001). However, first results from the southern tip of the São Francisco craton (D'Agrella-Filho et al., 2000) have revealed that even very-well preserved Neoproterozoic carbonate sequences, at the core of the craton, can be prone to remagnetization. The paleomagnetic poles obtained for these rocks fall at the Cambrian segment of the Gondwana apparent polar wander (APW) path, i.e. the carbonates have acquired their characteristic magnetization after the peak of the Brasiliano (Panafrican) deformation. Since this orogenic event affects the same sedimentary succession at the margin of the craton, the remanence in these rocks is logically secondary in origin. Moreover, these rocks share the same anomalous magnetic properties reported for other remagnetized carbonates, such as wasp-waisted hysteresis loops, anomalously high hysteresis ratios and contradictory Lowrie–Fuller and Cisowski tests (Jackson, 1990; Jackson et al., 1992; Channell and McCabe, 1994; Huang and Opdyke, 1996).

Continental to global-scale remagnetization events have been extensively reported for the Late Paleozoic and Cretaceous and are usually associated in space and time to the development of orogenic belts (e.g. McCabe and Elmore, 1989; Jackson et al., 1992). The interaction of sediments with fluids expelled or derived from the orogenic belts is the most common mechanism attributed for such widespread remagnetization events (Oliver, 1986; Garven, 1995). In order to test the continental-scale character of the Cambrian remagnetization in South America and its relation to the Brasiliano orogeny, a paleomagnetic, rock magnetic and geochronological survey has been performed in the Salitre Formation carbonates, eastern São Francisco craton. These sediments occur in the Irecê basin, about 600 km away from the previously studied area (Fig. 1). Also discussed are the mechanisms that could have promoted the simultaneous resetting of the isotopic and magnetic record and the implications of the new poles on the APW path of Gondwana.

2. Geological setting and sampling

The São Francisco and Irecê basins cover more than 300,000 km^2 of the São Francisco craton (Fig. 1). They comprise a basal glacial unit, covered by a thick carbonate sequence (Karfunkel and Hoppe, 1988). In the southwestern part of the São Francisco craton, along the São Francisco basin, these units are known as Macaúbas/Jequitaí (glacial) and Bambuí (carbonates) Groups. Along the Irecê basin they are named Bebedouro (glacial) and Salitre (carbonates) Formations. These successions are considered to be coeval and both host Pb–Zn deposits (Dardenne, 1979). Moderate to weak deformation is recorded inside the basins. It is a far-field response to the intense deformation that took place along the encircling Brasiliano fold belts. This deformation produced gentle folding in the center of the Irecê basin which grades northwards to tight folding and thrusts close to the Riacho do Pontal fold belt (Inda and Barbosa, 1978).

In the Irecê basin, the carbonates that directly overlie the diamictites and tillites, are white dolostones and pink to red argillaceous limestones. They show negative δ^{13}C values (mean at −5.1‰; Torquato and Misi, 1977) and sedimentary features such as columnar stromatolites and tepee-like structures (Misi and Kyle, 1994), which are typical of cap carbonates described elsewhere in the world (Kennedy, 1996; James et al., 2001). Dark-gray laminated limestones, which grade upward into gray to red dolostones and shales, and black oolitic limestones comprise two cycles of carbonate and pelitic-psamitic sedimentation. The values

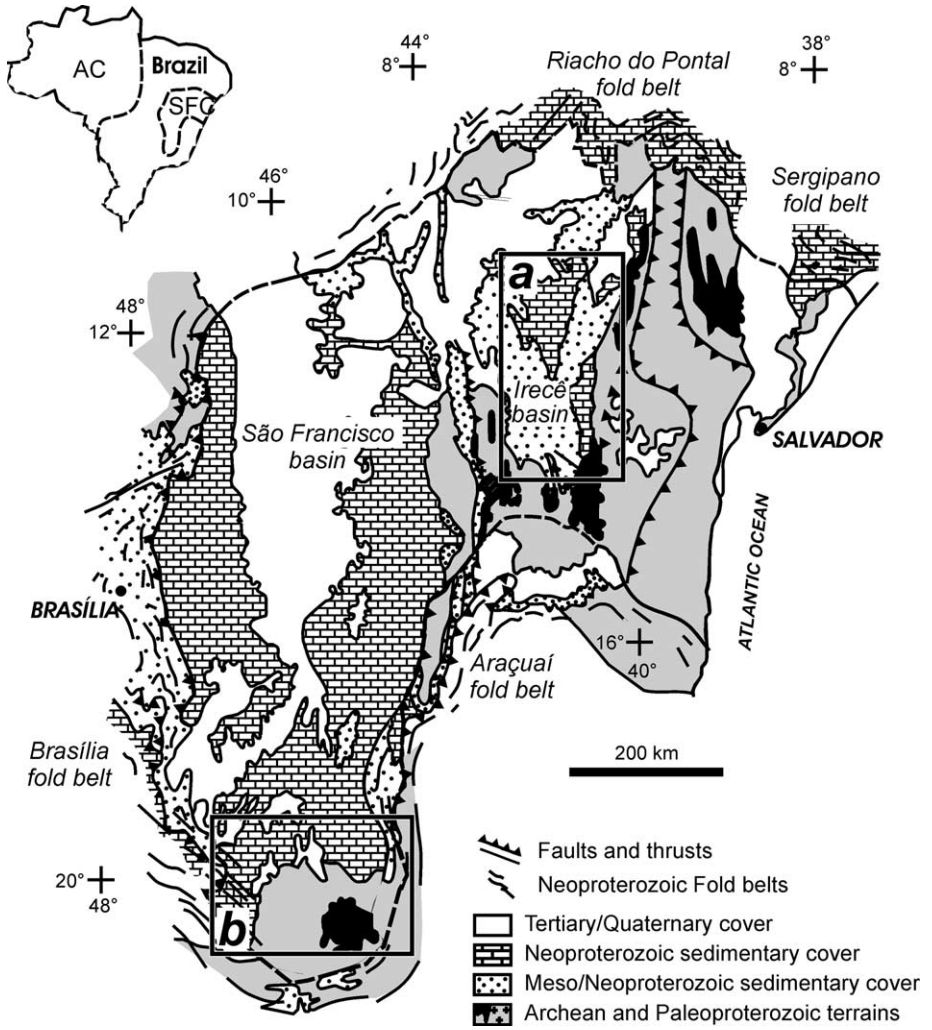


Fig. 1. Geological map of the São Francisco craton with indication of Neoproterozoic basins and surrounding fold belts (modified from Santos et al., 2000). Rectangles indicate the location of (a) Salitre and (b) Bambuí areas. The inset map shows the Amazon (AC) and the São Francisco (SFC) cratons in Brazil.

of $\delta^{13}C$ increase upwards, with a mean value of +8.53‰ at the top of the sequence (Misi and Veizer, 1998).

Paleomagnetic samples were collected at 36 nearly horizontal beds (less than 4° of inclination) from 13 outcrops and quarries along the best preserved sectors of the Irecê basin (A–M, in Fig. 2). Each sampled layer was considered as a sampling site. A first set of samples was collected in 1994, and preliminary results were presented by D'Agrella-Filho (1995). An additional field trip in 1999 complemented the collection. Four outcrops (BA98-15, -16, -23, and -28) were sampled for geochronological purposes (Fig. 2), with a total of 32 carbonate samples. Samples from BA98-15 outcrop comprise undeformed gray, strongly silicified carbonates with columnar stromatolites. Undeformed pink carbonates were collected at BA98-16 outcrop, just above diamictites from the Bebedouro Formation. These two outcrops represent the typical cap carbonate. The other samples are dark-gray carbonates from the middle-upper part of the Salitre Formation.

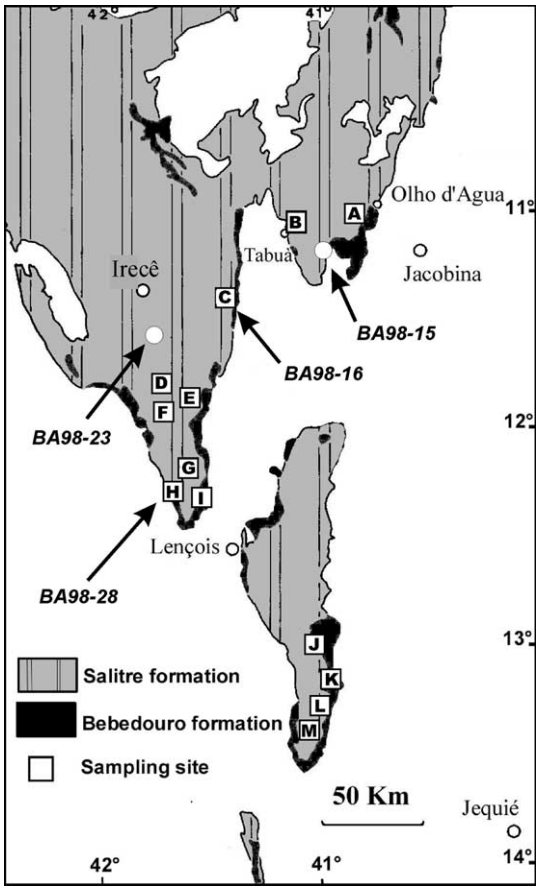


Fig. 2. Geological sketch map of the Irecê basin (after Karfunkel and Hoppe, 1988) with indication of paleomagnetic (A–M) and geochronological (BA98-15, -16, -23, and -28) sampling points. Location of the area is indicated by the rectangle (a) in Fig. 1.

3. Age constraints

3.1. Previous work

The ages of the São Francisco Supergroup units (glacial sediments and carbonates) are still arguable since various dating methods have shown contrasting results (see review in Babinski et al., 1999). A maximum sedimentation age of 906 ± 2 Ma for the glacial rocks is given by U–Pb data (zircon and badeleyte) for dikes that cut across the Mesoproterozoic Espinhaço Supergroup but do not intrude the Macaúbas Group sediments (Machado et al., 1989). This inference is in agreement with the analysis of detrital zircons from the glacial deposits, all of them older than 900 Ma (Pb–Pb data of Buchwaldt et al., 1999; and U–Pb SHRIMP ages of Pedrosa-Soares et al., 2000).

Most of the available geochronological data for the carbonate sequences were obtained for the southern part of the São Francisco basin (the Bambuí Group). Rb–Sr ages on clays and whole-rock samples range from 695 ± 12 Ma (R_0 = 0.7077) to 560 ± 40 Ma (R_0 = 0.7110) and K–Ar ages on fine fraction clays range from 662 ± 18 to ∼478 Ma (see compilation in Thomaz-Filho et al., 1998). The older ages were considered as minimum depositional ages, and the younger ones were interpreted as related to the influence of later thermal events in the margin of the craton at the end of the Brasiliano orogeny. Most of the Pb–Pb and U–Pb ages obtained for the basal units of the Bambuí Group are coeval or younger than the tectonic activity on the marginal fold belts, and range from 550 to 500 Ma (Babinski et al., 1999). Only two ^{207}Pb–^{206}Pb isochron ages from the basal unit of the Bambuí Group are older than 600 Ma, and may thus represent the time of deposition. An age of 686 ± 69 Ma was obtained by Babinski et al. (1999) and gives a minimum depositional age for the carbonates. Interestingly, this age overlaps the ages recently obtained on the Windermere Supergroup by Lund et al. (2003) and interpreted by those authors as the age of the Rapitan glaciation. Another date of 762 ± 29 Ma was recently obtained by Babinski et al. (2002) on well preserved cap carbonates from the southern part of the basin (see Peryt et al., 1990). These ages fall close to the ages classically attributed to the Sturtian glacial interval (Evans, 2000), and are interpreted as the age of deposition of these rocks. New geochronological data for carbonate rocks from the Irecê basin (Salitre Formation) are reported below.

3.2. Pb–Pb dating of Salitre carbonates

Pb isotopic analyses were carried out at the Geochronological Research Center of University of São Paulo following the procedures described in Babinski et al. (1999). Pb concentrations were obtained by the isotope dilution mass spectrometry (IDMS) technique using ^{208}Pb spike. Isochron ages were calculated using the Isoplot software of Ludwig (1999), and the errors are reported as 1 sigma.

Table 1
Pb isotope data from Salitre Formation carbonates, São Francisco craton

Sample	$^{206}Pb/^{204}Pb$	1σ	$^{207}Pb/^{204}Pb$	1σ	$^{208}Pb/^{204}Pb$	1σ	Pb (ppm)
BA98-15.1	51.459	0.322	17.717	0.247	41.660	0.245	
BA98-15.1	51.014	0.160	17.599	0.160	41.420	0.110	0.23
BA98-15.1	58.900	0.121	18.094	0.119	42.260	0.122	
BA98-15.2	102.348	0.159	20.650	0.140	48.339	0.145	
BA98-15.2	104.223	0.030	20.684	0.030	48.147	0.030	0.13
BA98-15.2	100.941	0.140	20.490	0.124	47.210	0.131	
BA98-15.3	100.144	0.142	20.460	0.049	47.602	0.048	0.14
BA98-15.4	63.140	0.132	18.397	0.140	47.073	0.140	
BA98-15.4	65.298	0.050	18.539	0.040	47.449	0.050	0.30
BA98-15.4	65.960	0.195	18.566	0.177	47.404	0.179	
BA98-15.5	47.020	0.196	17.418	0.260	44.044	0.261	0.32
BA98-15.5	45.828	0.018	17.320	0.018	43.923	0.017	0.32
BA98-15.5	46.560	0.036	17.406	0.036	43.920	0.036	
BA98-15.6	43.968	0.048	17.347	0.064	45.027	0.064	
BA98-15.6	44.604	0.158	17.394	0.153	44.968	0.158	
BA98-15.7	48.346	0.114	17.542	0.119	45.834	0.119	
BA98-15.7	61.195	0.080	18.363	0.080	50.364	0.080	0.37
BA98-15.7	48.941	0.038	17.648	0.035	46.143	0.038	
BA98-15.8	60.645	0.065	18.316	0.070	49.780	0.070	
BA98-15.8	62.729	0.093	18.354	0.092	49.787	0.091	
BA98-15.9	52.547	0.026	17.740	0.030	42.999	0.031	0.53
BA98-15.10	31.000	0.040	16.558	0.040	41.670	0.040	0.73
BA98-15.10	30.142	0.058	16.499	0.057	41.271	0.060	
BA98-15.11	41.733	0.075	17.155	0.072	42.897	0.075	
BA98-15.12	35.951	0.030	16.795	0.030	41.163	0.030	0.39
BA98-15.12	36.300	0.076	16.778	0.073	40.914	0.077	
BA98/02-01	27.608	0.056	16.254	0.055	40.607	0.053	
BA98/02-02	26.688	0.036	16.204	0.035	41.821	0.036	
BA98/02-03	27.533	0.039	16.312	0.035	42.209	0.039	
BA98/02-04	28.940	0.035	16.369	0.035	41.108	0.036	
BA98/02-05	31.602	0.058	16.541	0.056	41.530	0.061	
BA98-28A L2	19.948	0.012	16.004	0.011	40.191	0.012	6.71
BA98-28A L3	19.867	0.011	15.988	0.011	40.172	0.011	7.86
BA98-28C L2	19.502	0.037	15.980	0.036	39.959	0.037	
BA98-28C L3	19.380	0.011	16.037	0.012	39.805	0.014	
BA98-28D L2	20.106	0.022	16.003	0.021	40.097	0.021	
BA98-28D L3	20.047	0.013	15.975	0.012	40.084	0.013	
BA98-28E L2	20.808	0.028	16.016	0.023	41.143	0.030	2.64
BA98-28E L3	21.240	0.034	16.079	0.034	41.998	0.035	1.82
BA98-28Fe L2	21.134	0.018	16.074	0.018	41.074	0.019	
BA98-28Fe L3	21.446	0.008	16.067	0.009	42.167	0.009	
BA98-28Fc L2	20.092	0.084	15.972	0.061	39.502	0.082	
BA98-28Fc L3	20.042	0.010	16.016	0.008	39.689	0.009	
BA98-28H L2	21.845	0.047	15.997	0.043	42.359	0.047	
BA98-28H L3	22.203	0.014	16.122	0.014	42.932	0.015	
BA98-28G L2	20.195	0.020	15.958	0.017	40.644	0.017	
BA98-28G L3	20.075	0.013	15.968	0.013	40.648	0.007	
BA98-16E	19.861	0.103	15.767	0.102	41.700	0.106	
BA98-16F	19.797	0.020	15.892	0.022	40.415	0.023	
BA98-16G	20.083	0.008	15.895	0.008	40.971	0.008	
BA98-16H	20.487	0.021	15.899	0.018	41.509	0.023	
BA98-16I	19.643	0.028	15.828	0.028	40.084	0.030	

Table 1 (*Continued*)

Sample	$^{206}Pb/^{204}Pb$	1σ	$^{207}Pb/^{204}Pb$	1σ	$^{208}Pb/^{204}Pb$	1σ	Pb (ppm)
BA98-16J	20.179	0.022	15.938	0.019	40.729	0.019	
BA98-16K	19.924	0.026	15.923	0.024	40.405	0.025	
BA98-16L	20.525	0.007	15.934	0.007	40.423	0.007	
BA98-23A	22.183	0.018	15.943	0.018	40.150	0.019	
BA98-23B	22.866	0.020	16.023	0.019	40.288	0.023	
BA98-23C	24.592	0.005	16.134	0.005	41.239	0.005	
BA98-23D	24.136	0.004	16.088	0.004	40.663	0.004	

L1 was discarded for all samples and only L2 was analyzed (see Babinski et al., 1999 for analytical details). For BA98-28 samples an additional leach was performed because they have a pelitic component and did not dissolve like the other samples. Isotopic compositions were corrected for a mass fractionation factor of 0.12%/amu, determined through more than 200 measurements of the Common Pb Standard NBS 981. Concentrations were determined using ^{208}Pb spike. Analytical blanks range from 30 to 50 pg and have negligible affect on the measured Pb isotopic compositions.

Fifty-nine lead isotopic analyses were obtained on 37 samples collected from four different outcrops, and the results are presented in Table 1.

Seventeen samples of stromatolitic carbonates from the BA98-15 outcrop were analyzed in duplicate or triplicate. Their Pb isotopic compositions are very radiogenic, with $^{206}Pb/^{204}Pb$ ratios ranging from 26.7 to 104.2, $^{207}Pb/^{204}Pb$ from 16.20 to 20.68, and $^{208}Pb/^{204}Pb$ from 40.5 to 50.4. Pb concentrations in these carbonates are very low, varying between 0.13 and 0.73. The lowest concentrations were determined on samples which have the most radiogenic Pb ratios, as already observed by Babinski et al. (1995, 1999). In the Pb isochron diagram these ratios yield a $^{207}Pb/^{206}Pb$ age of 514 ± 33 Ma (Fig. 3a).

In contrast to the stromatolitic carbonates, samples from the other localities presented almost non-radiogenic to non-radiogenic Pb isotopic ratios and relatively high Pb concentrations. The limited range of the Pb ratios does not allow to define isochron ages for these sites (Fig. 3b). Eight pink cap carbonates (BA98-16; Table 1) showed Pb isotopic compositions with $^{206}Pb/^{204}Pb$ ratios ranging from 19.6 to 20.5, $^{207}Pb/^{204}Pb$ from 15.77 to 15.94, and $^{208}Pb/^{204}Pb$ from 40.1 to 41.7. Similarly, four samples from the gray undeformed carbonates (BA98-23; Table 1) yielded $^{206}Pb/^{204}Pb$ ratios ranging from 22.2 to 24.6, $^{207}Pb/^{204}Pb$ from 15.94 to 16.13, and $^{208}Pb/^{204}Pb$ from 40.2 to 41.2. No Pb concentrations were determined on these samples.

Pb analyses of dark-gray deformed pelitic carbonates (BA98-28) were done using an additional leachate (0.7N Hbr) (see details in Babinski et al., 1999). The second leachate (L2) used for the other samples did not dissolve most of the samples from this outcrop due to their pelitic component. These samples also showed non-radiogenic Pb ratios and high Pb concentrations (Table 1), between 1.82 and 7.86 ppm. The $^{206}Pb/^{204}Pb$ ratios range from 19.4 to 21.8, $^{207}Pb/^{204}Pb$ from 15.96 to 16.08, and $^{208}Pb/^{204}Pb$ from 39.5 to 42.9.

Although these non-radiogenic isotopic ratios are scattered (Fig. 3b), they fall at the lower end of the line defined by the ratios of stromatolitic cap carbonates (BA98-15) (Fig. 3a). The Pb–Pb age obtained considering the ratios of BA98-15 plus BA98-16 samples is 517 ± 27 Ma. A similar age of 522 ± 25 Ma is also obtained with the incorporation of the BA98-23 samples. Finally, a regression of all Pb ratios, even those of the BA98-28 outcrop, which could have an older pelitic component, gives an age within the same range (486 ± 28 Ma), considering the analytical error.

4. Paleomagnetism

4.1. Experimental procedure

Paleomagnetic samples were oriented in the field by sun and magnetic compasses. In the laboratory, samples were cut into 457 cylindrical specimens (2.2 cm × 2.5 cm) and submitted to stepwise thermal and AF treatments. Samples collected in the 1999 field campaign were stored in a low-field magnetic chamber for 2 months before measurements. Paleomagnetic analysis were performed in the Paleomagnetic Lab of

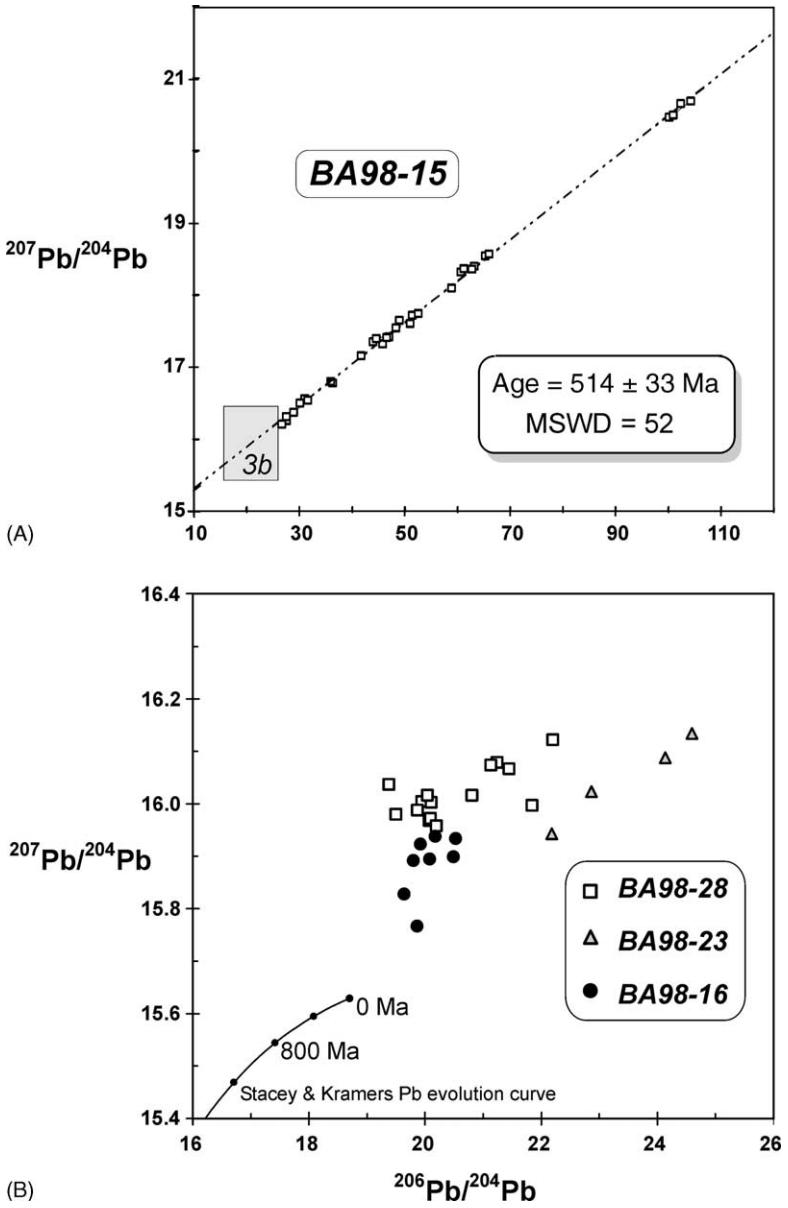


Fig. 3. (a) Pb isochron diagram for stromatolitic carbonates of site BA98-15. The range of values represented in (b) is indicated by a gray rectangle. (b) Pb diagram for the other sites (BA98-16, -23, and -28). The Stacey and Kramers's (1975) Pb evolution curve is given for reference.

University of São Paulo. Remanent magnetizations were measured with a 2G-cryogenic magnetometer. Alternating field (AF) demagnetizations were done by an automated three-axis AF-demagnetizer coupled with the magnetometer. Thermal demagnetizations were performed with a Magnetic Measurements oven (MMTD60; peak temperatures within $\pm 2\,^{\circ}C$). These instruments are housed in a magnetically-shielded room with ambient field $<1000\,nT$. Magnetic components for each specimen were identified in

orthogonal plots, and calculated using the least squares fit method (Kirschvink, 1980). Fisher's (1953) statistics was used to calculate vector mean directions and paleomagnetic poles.

4.2. Magnetic components

Thermal treatment was more efficient than AF demagnetization in isolating the magnetic components

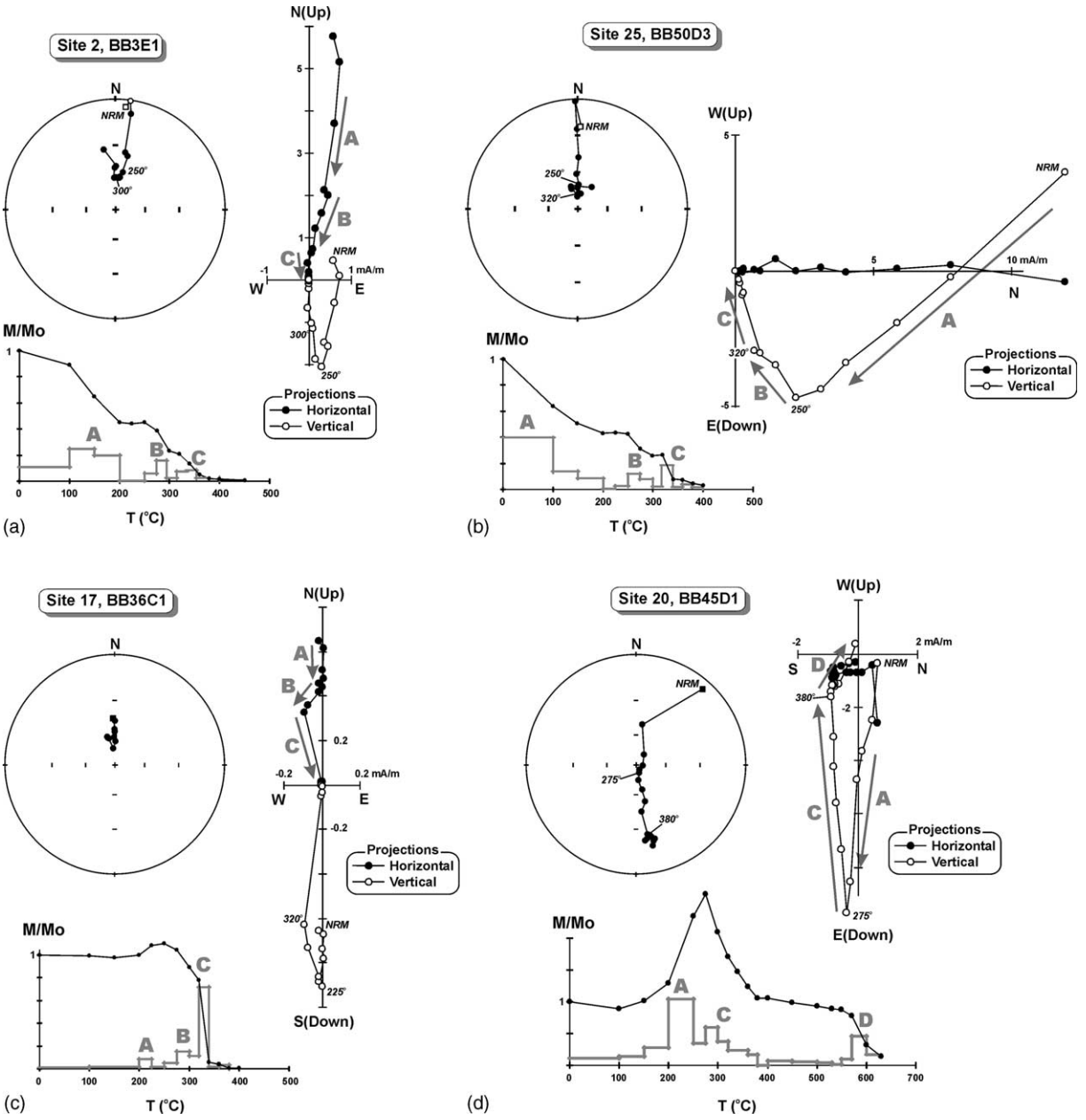


Fig. 4. Typical demagnetization trajectories after thermal treatment of Salitre samples. Magnetization intensity decay curves, orthogonal and stereographic projections are represented for each sample. Components A, B, C, and D are indicated by arrows in the orthogonal plots and correspond to peaks in the unblocking temperature spectra. Open (full) symbols correspond to negative (positive) inclinations in the stereographic projections.

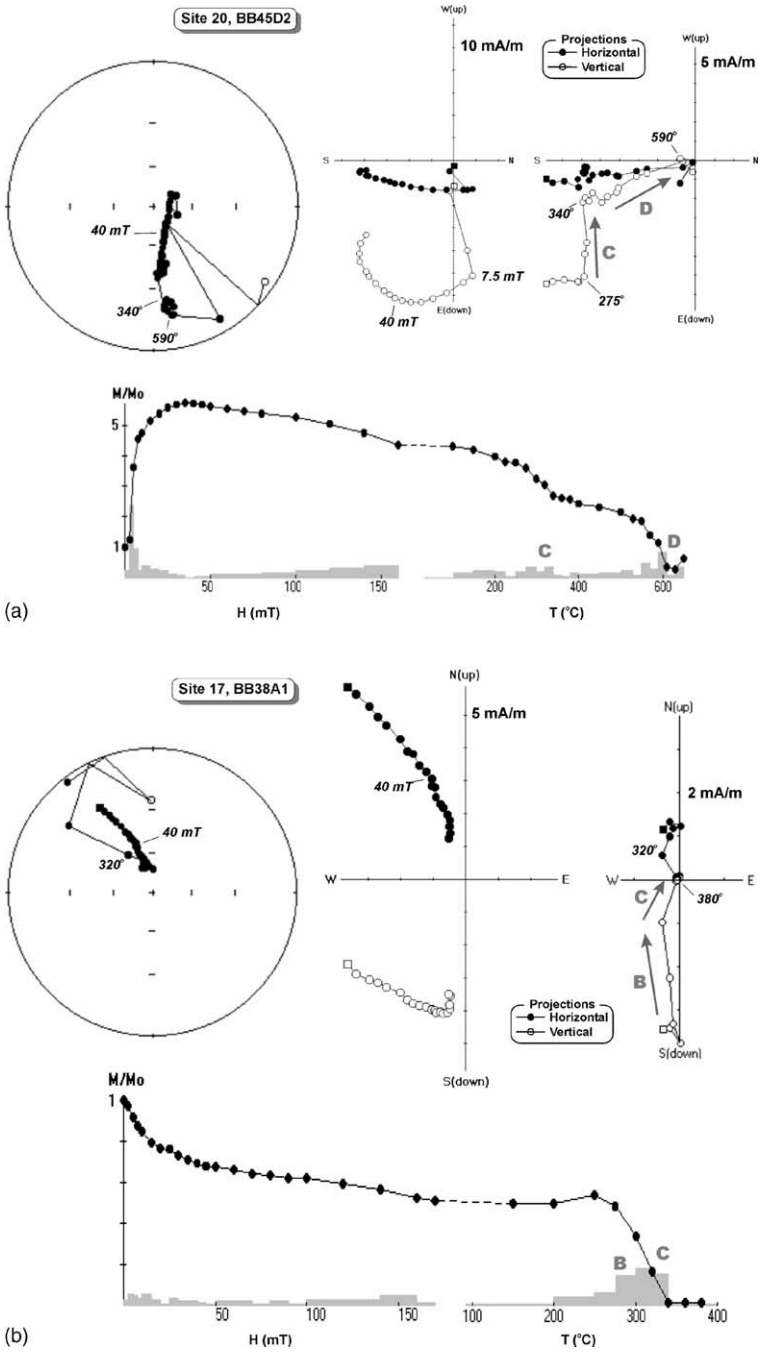


Fig. 5. Typical demagnetization trajectories after AF and thermal treatment of Salitre samples. Magnetization intensity decay curves, orthogonal and stereographic projections are represented for each sample. Components B, C, and D are indicated by arrows in the orthogonal plots and correspond to peaks in the coercivity spectra. Open (full) symbols correspond to negative (positive) inclinations in the stereographic projections.

Table 2
Components A, B, C, and D and virtual geomagnetic poles (VGP) for each site

Site	*n*/*N*	Characteristic remanent magnetization (ChRM)				VGP	
		D (°)	*I* (°)	α_{95} (°)	*k*	Plong. (°E)	Plat. (°N)
Component A							
02	4/12	7.1	−46.3	3.3	766.2	115.2	74.2
12	7/8	0.6	−55.7	5.1	140.5	137.3	66.2
13	7/19	355.0	−56.1	7.5	65.5	148.1	65.4
14	6/12	5.4	−57.4	3.8	306.2	128.7	64.0
15	5/10	359.1	−57.7	7.2	113.7	140.1	64.1
16	7/10	355.9	−54.6	8.2	55.5	147.1	67.0
17	8/12	0.9	−61.5	4.9	128.2	137.2	59.8
20	5/15	0.4	−67.3	13.3	34.2	137.9	51.7
23	5/8	1.0	−48.3	8.0	92.4	135.5	72.6
24	7/10	0.6	−47.1	6.5	87.7	136.5	73.6
25	4/11	8.2	−45.3	6.2	219.0	113.2	72.8
48	6/8	330.4	−50.4	13.9	24.3	188.3	56.2
50	4/12	332.6	−44.6	8.3	123.1	194.9	60.1
55	7/13	35.5	−47.7	12.3	25.2	81.5	52.6
Component B							
01	4/10	25.6	51.5	13.0	50.8	346.9	38.3
02	3/12	17.6	59.0	8.2	227.0	335.4	34.4
13	6/19	76.9	69.5	10.0	45.5	354.2	−2.3
14	6/12	55.7	77.6	6.6	102.8	337.8	1.3
15	5/10	58.1	70.7	14.6	28.2	347.8	6.9
16	4/10	44.4	73.5	13.4	48.2	339.6	9.8
17	8/12	33.5	69.7	5.3	108.8	338.6	18.1
21	6/10	36.8	58.0	10.1	44.7	350.6	28.9
27	3/11	17.9	74.6	13.3	87.3	327.5	16.0
37	10/11	43.9	44.5	12.5	15.9	4.1	36.0
40	5/7	35.0	47.3	14.5	29.0	358.6	37.8
46	5/8	25.7	46.9	10.3	55.8	350.7	43.4
57	6/12	13.5	42.8	6.0	124.7	338.5	51.4
Component C							
01	7/10	349.2	63.7	6.4	90.6	310.2	30.6
02	8/12	1.5	63.4	5.0	123.2	320.3	31.7
12	8/8	0.0	63.9	3.0	353.1	318.5	31.9
13	14/19	6.9	72.3	4.0	99.0	322.4	19.8
14	12/12	7.6	67.7	3.5	156.3	323.8	26.5
15	10/10	6.8	68.9	5.7	65.1	323.0	24.9
16	10/10	7.5	65.6	4.8	101.6	324.3	29.4
17	12/12	354.0	67.4	2.8	249.7	314.1	27.1
20	12/15	356.5	74.5	4.7	86.6	316.5	17.1
22	15/21	6.1	61.8	2.5	229.0	323.7	34.9
23	8/8	347.6	64.4	5.3	111.7	308.3	30.8
24	8/10	350.2	73.1	5.1	119.7	313.0	19.0
25	10/11	358.7	66.7	6.0	65.8	317.6	29.3
26	9/10	8.3	63.2	5.8	78.8	325.7	33.3
46	8/8	9.9	56.0	8.0	48.9	329.5	41.3
47	7/8	10.0	64.8	7.3	70.3	326.9	31.4
48	4/8	5.9	60.0	10.3	81.3	324.5	37.5
49	12/12	11.4	57.9	6.6	44.8	330.4	39.1
50	9/12	0.8	55.5	8.5	37.6	319.8	42.7
52	10/12	5.3	52.3	10.0	24.4	325.3	45.6

Table 2 (*Continued*)

Site	*n*/*N*	Characteristic remanent magnetization (ChRM)				VGP	
		D (°)	*I* (°)	α_{95} (°)	*k*	Plong. (°E)	Plat. (°N)
54	11/12	36.7	54.7	6.1	57.1	353.7	31.8
55	10/13	8.0	57.6	5.6	74.9	326.8	39.8
56	7/8	6.1	60.1	6.0	102.4	324.4	37.3
57	12/12	19.1	51.7	4.3	101.1	340.7	42.5
Component D							
20	4/15	170.1	27.7	5.6	269.7	30.6	−80.0

The mean ChRM is given by its declination (*D*), inclination (*I*), radius of the 95% confidence cone (α_{95}), and the precision parameter (*k*) (Fisher, 1953). *n*/*N*: number of samples considered for the mean/number of analyzed samples.

of the samples (Figs. 4 and 5). One to four magnetic components were identified for each analyzed specimen (Fig. 4 and Table 2). Some specimens have shown a magnetic component very close to the present geomagnetic field up to 150 °C heating. This component is interpreted as a viscous magnetization, which was not eliminated even after storage in the magnetically-shielded room. In fourteen sites, a northerly, steep negative direction (component A) was disclosed after heating between 100–150 and 275 °C (Fig. 4a–d), or after the AF demagnetization at 25 mT. Two slightly different, north to northeast, positive inclination directions were further isolated at higher alternating fields and unblocking temperatures (Fig. 4a–c). The component B was isolated in thirteen sites after heating between 250–275 and 320 °C. The component C was isolated in twenty-four sites after heating between 300–340 and 550 °C. AF treatment up to 160 mT was not effective in completely demagnetizing most of the samples, and thus failed in isolating components B and C. The same samples were further completely demagnetized by thermal treatment, isolating components B and C (Fig. 5). This indicates that these components reside in hard coercivity fractions. Site-mean directions for components A, B and C are presented in Fig. 6. After heating above 550 °C all sites but one (site 20, pink cap carbonate) have shown an erratic behavior. In site 20 an additional, positive and southward-directed component D was disclosed after heating between 530 and 630 °C (Figs. 4d and 5a), giving a site-mean direction of Dec = 170.1°, Inc = +27.7° ($k = 269.7$, $\alpha_{95} = 5.6°$).

5. Magnetic mineralogy

5.1. Remanence carriers

Data from AF and thermal treatments suggest three magnetic phases, namely monoclinic pyrrhotite, magnetite and hematite, as the carriers of the more stable B, C, and D magnetic components in the Salitre rocks.

Several samples have shown the component B, with unblocking temperatures in the 250–340 °C interval (Fig. 4a–c) which is compatible with monoclinic pyrrhotite (Rochette et al., 1990). Most of these samples also show a strong increase in magnetic susceptibility during thermal treatment at higher temperatures, which is also typical of thermally induced sulphide alteration (Dekkers, 1990). In contrast, temperatures as high as 550 °C, close to the Curie temperature of magnetite, were necessary to unblock the component C for most of the sites (Fig. 4a–d). In some sites, however, the component C has been completely demagnetized at 320–340 °C (Figs. 4b and 5a) suggesting that pyrrhotite may also be a carrier of this component. In addition to magnetite and sulphides, the pink to red cap carbonates probably have some amount of hematite, as suggested by thermal unblocking temperatures above 600 ° C in hard fractions, resistant to AF demagnetization up to 160 mT (Fig. 5a). However, only samples from site 20 have shown a stable direction associated to such high unblocking temperatures (component D).

Thermal demagnetization of a tri-orthogonal IRM acquired at 1.3, 0.3 and 0.1 T fields (Lowrie, 1990) was performed in four selected samples. Results for three

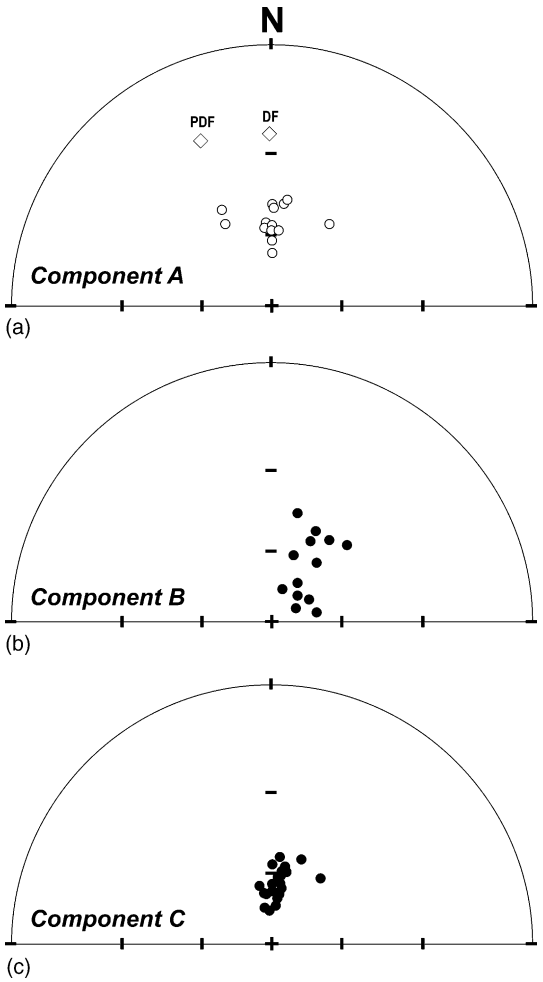


Fig. 6. Site-mean directions for components A, B, and C. Open (full) symbols correspond to negative (positive) inclinations. PDF: present day geomagnetic field; DF: dipolar field.

samples from sites 2, 16 and 20 are shown in Fig. 7. Two of these sites (sites 2 and 16) presented the components B and C, while site 20 does not show the component B but the components C and D (Figs. 4d and 5a). In all samples the soft (<0.1 T) fraction was gradually demagnetized up to 500–550 °C, showing the long demagnetization tails that typify MD magnetite (Dunlop and Özdemir, 2000). For samples of sites 2 and 16 the medium (0.1–0.3 T) and hard (>1.3 T) fractions show a strong decrease of magnetization at 300–340 °C, suggesting that pyrrhotite is a major carrier of remanence in these rocks, but the medium fraction is completely demagnetized only at around 500 °C. In contrast, the hard and medium fractions of

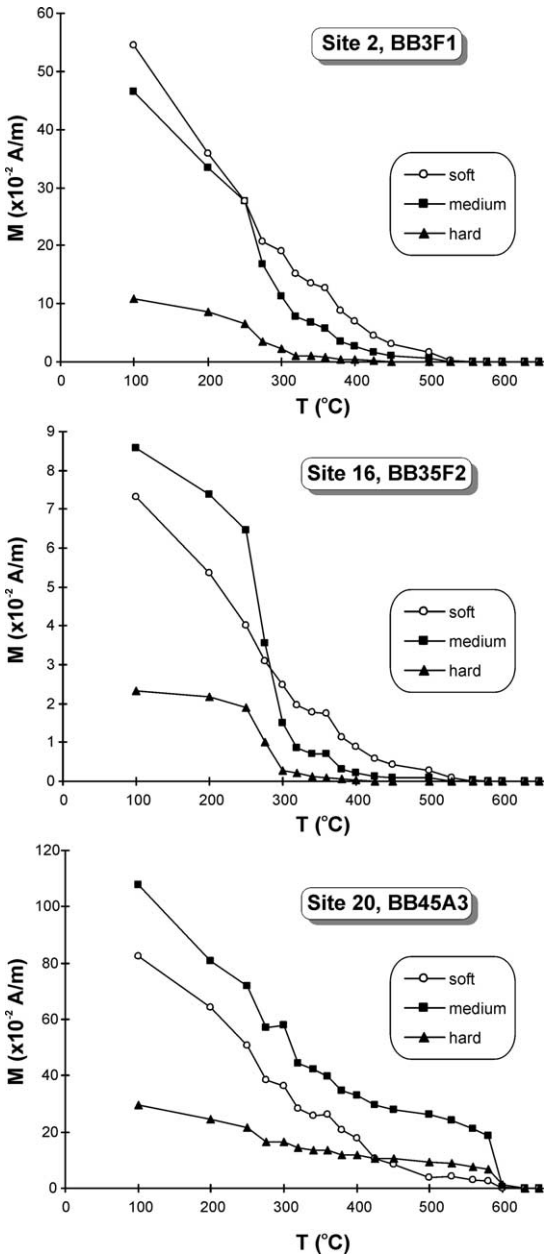


Fig. 7. Examples of thermal demagnetization of a tri-orthogonal IRM (soft: 0.1 T, medium: 0.3 T, hard: 1.3 T).

site 20 do not show a drop of magnetization at around 300–340 °C. Instead, these fractions are associated to high unblocking temperatures, in excess of 600 °C for the hard component, confirming the presence of magnetite and hematite in this sample.

5.2. Rock magnetic fingerprints of remagnetization

Hysteresis parameters have been used as a fingerprint of remagnetization in carbonate rocks worldwide (Jackson, 1990; Channell and McCabe, 1994; Tarduno and Myers, 1994). Hysteresis loops were performed for samples from nineteen sampling sites, within which most presented both components B and C (or only the later), one presented the components C and D (sample BB45E, site 20) and another presented the component B only (sample BB122A, site 37). Almost all these samples also have the component A. For all samples hysteresis curves are wasp-waisted in shape. This behavior is typical of a mixture of soft and hard coercivity fractions, and has been found in all remagnetized carbonates to date. Accordingly, all samples show anomalously high H_{cr}/H_c ratios when plotted in the Day et al. (1977) diagram (Fig. 8). The Salitre samples with component C follow a power-law fit of $Mrs/Ms = 0.98(H_{cr}/H_c)^{-0.8}$ and align just below the trend of remagnetized carbonates proposed by Jackson (1990). This trend has been interpreted as a result of the mixing of stable SD magnetite grains and very fine SP particles, with grain-sizes in the range of 9–12 nm (Dunlop, 2002). The other two samples fall above the "remagnetization line". Sample BB45E has small H_{cr}/H_c values and falls at the tip of the regression line defined by samples with component C. Sample BD122A presents a Mrs/Ms value slightly higher than 0.5, above the "SP saturation envelope" modeled by Dunlop (2002) for the Day plot, but within the characteristic range for monoclinic pyrrhotite (Rochette et al., 1990).

In addition to the hysteresis curves, the modified form of the Lowrie–Fuller test (Johnson et al., 1975) and the Cisowski (1981) test were performed on thirteen representative samples of Salitre carbonates. Samples were initially demagnetized in alternating field (AF) at 200 mT, and then given an anhysteretic remanent magnetization (ARM) in a peak field of 200 mT and a biasing field of 0.1 mT. The ARM was then stepwise demagnetized in AF. Finally, samples were given an incremental isothermal remanent magnetization (IRM) up to 200 mT and were again stepwise demagnetized in AF.

The resulting curves for three representative samples are shown in Fig. 9. The results for samples with component C (Fig. 9a) is similar to the contradictory behavior observed in other remagnetized successions (Jackson, 1990; McCabe and Channell, 1994; Huang and Opdyke, 1996). The symmetrical IRM acquisition and demagnetization curves, crossing

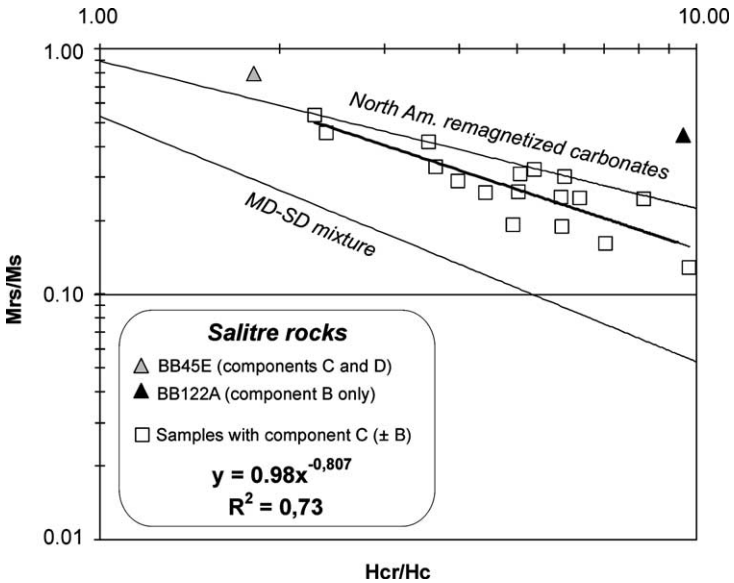


Fig. 8. Day plot for Salitre carbonate samples. A regression line is given for samples with component C (some of them also showing component B).

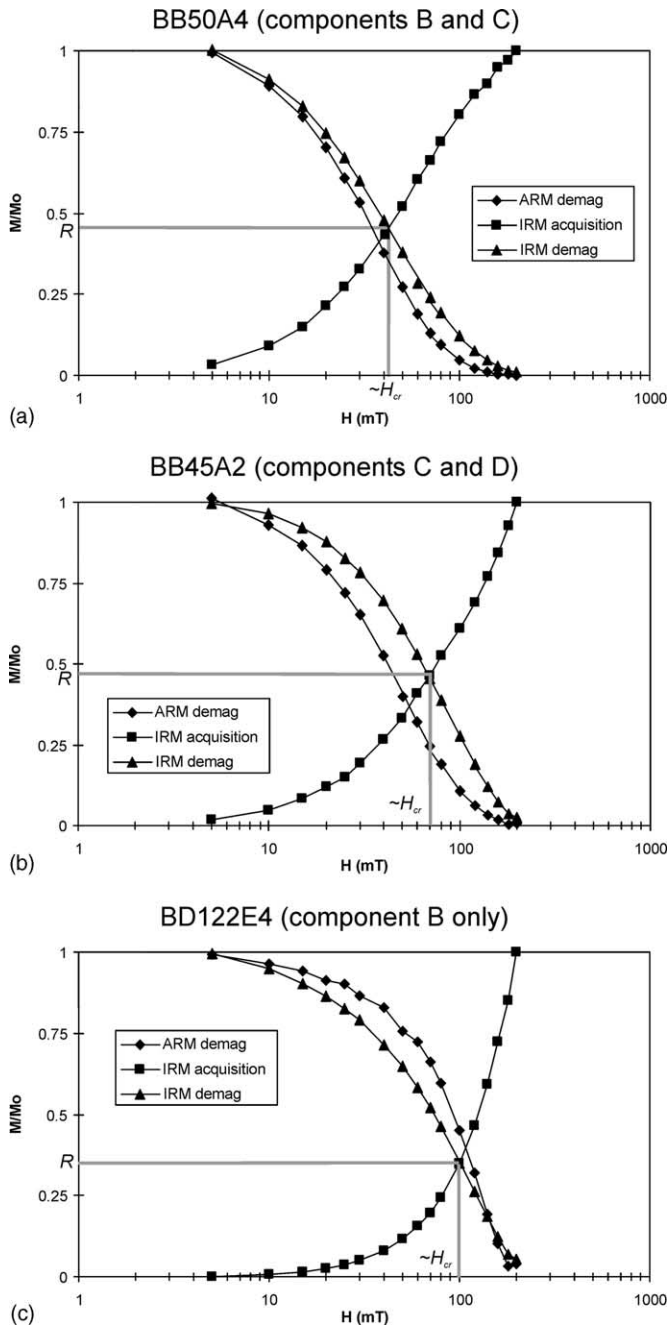


Fig. 9. Results of Lowrie–Fuller–Cisowski test. H_{cr}: remanent coercivity, R: crossing point of IRM acquisition and demagnetization curves.

at around 50% are an indication that remanence is carried by non-interacting SD particles (Cisowski, 1981). The values of field crossing points for IRM acquisition and demagnetization ($\approx H_{rc}$) within the 50–70 mT range (Symons and Cioppa, 2000), and the upward-convex shape of the IRM-demag curve (Dunlop, 1983) are also typical of SD (to PSD) grains. At the same time, the ARM is weaker than IRM to AF

demagnetization in these samples. According to the modified Lowrie–Fuller test this behavior characterizes coarse-grained (MD) magnetite (Johnson et al., 1975; Dunlop, 1983). A similar result was obtained for sample BB45A2 (Fig. 9b), which presented components C and D, even though its IRM acquisition curve is far from saturation at 200 mT due to the influence of hematite. Samples with only component B have shown a different response to the tests (Fig. 9c). In contrast to the typical behavior of remagnetized carbonates, their ARM is systematically harder than the IRM to AF demagnetization. Moreover, their IRM acquisition and demagnetization plots are not symmetrical. Instead, they mimic the standard cross-over plots for SD-PSD pyrrhotite of Symons and Cioppa (2000), crossing at around 100 mT, well above the magnetite upper limit given by those authors.

6. Discussion

6.1. Remagnetizations across the São Francisco and Irecê basins

Mean directions and paleomagnetic poles for the Neoproterozoic carbonates which cover the São Francisco craton are presented in Table 3. The paleomagnetic poles calculated from components A, B, and C of Salitre carbonates are similar to those obtained in the southern part of the São Francisco basin (Bambuí carbonates) by D'Agrella-Filho et al. (2000). The similarity among magnetic carriers and paleomagnetic poles is strong evidence for the common nature of the magnetizations in both areas.

Most of the Salitre samples show a multicomponent behavior (components A, B, C, and D), the magnetizations being carried by magnetite, pyrrhotite and hematite. Three of these components (A, B, and C) were also identified in the Bambuí rocks within the same intervals of unblocking temperatures (D'Agrella-Filho et al., 2000). The component A is interpreted as a soft magnetization acquired by MD magnetite. It is depicted in the soft fraction of all tri-orthogonal demagnetization plots (e.g. Fig. 7). The B component is assigned to pyrrhotite based on its high coercivities and characteristic unblocking temperatures (below 340 °C; Fig. 4a–c). Also, the analysis of Salitre samples for which only this component has been disclosed revealed high Mrs/Ms ratios (Fig. 8), and cross-over plots typical of SD-PSD pyrrhotite (Fig. 9c). These results contrast sharply to those obtained in samples with component C. The component C was unblocked at temperatures up to 550 °C at most

Table 3
Paleomagnetic poles for carbonates from Salitre Formation and Bambuí Group, São Francisco craton

Unit, component	N	Mean direction				Paleomagnetic pole				
		D_m (°)	I_m (°)	α_{95}	k	Lat. (N°)	Long. (°E)	α_{95} (°)	k	R
Bambuí, A (BaA)	31	358.0	−64.1	2.3	130.9	63.9	137.4	3.2	64.6	30.5
Bambuí, B (BaB)	33	25.6	68.3	1.6	247.5	14.7	330.7	2.5	104.3	32.7
Bambuí, C (BaC)	17	6.9	58.4	3.2	123.2	30.2	321.0	3.8	91.2	16.8
Salitre, A (SaA)	14	359.4	−53.9	5.8	47.3	67.2	139.7	7.3	30.6	13.6
Salitre, B (SaB)	13	34.0	61.4	7.8	29.0	25.0	345.1	10.2	17.3	12.3
Salitre, C (SaC)	24	5.8	63.2	2.9	105.1	32.6	323.1	4.0	23.6	23.6
Salitre + Bambuí mean pole − component A ($N = 45$)						64.9	138.1	3.1	48.6	44.1
Salitre + Bambuí mean pole − component B ($N = 46$)						–	–	–	32.4	44.6
Salitre + Bambuí mean pole − component C ($N = 41$)						31.6	322.2	2.8	65.3	40.4

N is the number of sites. The mean direction is given by its declination (D_m) and inclination (I_m), and the paleomagnetic pole by its latitude (Lat.) and longitude (Long.). R, α_{95} and k are the Fisher's (1953) statistical parameters. Bambuí poles are compiled from D'Agrella-Filho et al. (2000); the other poles were obtained in this work (results supersede preliminary ones presented in D'Agrella-Filho et al., 2000). The McFadden and Lowes (1981) test was performed for each component. The hypothesis that two populations of VGPs (N_1, R_1 and N_2, R_2), with similar precisions (k) share a common true mean pole may be rejected at the level of confidence P, if $U > (1/P)^{1/(N-2)} - 1$ ($P = 0.05$). $U = [R_1 + R_2 - R_2/(R + R_2)]/2(N - R_1 - R_2)$.

of the sites. This and the results from hysteresis loops and tri-orthogonal demagnetization indicate that this component resides dominantly in stable SD magnetite. At some sites, however, the component C may be carried by pyrrhotite as suggested by unblocking temperatures around 300–320 °C. It is worth noting that samples for which the component C was isolated, in spite of having or not an additional component carried by pyrrhotite or hematite, have systematically presented the typical rock magnetic signature of remagnetized carbonates, i.e. wasp-waisted hysteresis loops, contradictory Lowrie–Fuller and Cisowski tests, and anomalously high hysteresis ratios following the "remagnetization trend" of Jackson (1990) in the Day plot.

The Bambuí and Salitre components give rise to similar paleomagnetic poles. Poles A (SaA, BaA) and C (SaC, BaC) have overlapping confidence circles, while poles B (SaB, BaB) plot close but do not coincide. Note, however, that component B was identified at comparatively fewer sites and is more scattered in Salitre than in Bambuí. In order to check the similarity of the Salitre and Bambuí poles, we have applied the McFadden and Lowes (1981) test, which discriminates mean directions obtained from Fisher populations. Since the collections to be compared are 600 km apart, we have applied the test to their virtual geomagnetic poles instead of their mean-site directions (see Table 3). According to the test, the poles obtained from components A and C for both collections are statistically undistinguishable at 95% of confidence, while poles derived from component B are statistically different. Mean paleomagnetic poles A and C for the whole set of Salitre and Bambuí PGVs were then calculated (Table 3).

Following the interpretation of D'Agrella-Filho et al. (2000), component A (Fig. 6a) is tentatively attributed to a Paleozoic thermoviscous remagnetization event, based on the similarity of its paleomagnetic pole to other Phanerozoic poles of South America (Rapalini and Tarling, 1993). The components B and C (Fig. 6b–c) are associated to higher unblocking temperatures and remanent coercivities, and are interpreted as a more stable record of the ancient Earth field. These components present a single magnetic polarity over the whole sedimentary succession. The directions show a tight grouping within each component as well. Since peak temperatures within the central part of the basin are less than 200 °C (Babinski et al., 1989), these components are interpreted as post-depositional chemical remanent magnetizations (CRM), acquired almost simultaneously across the whole sedimentary pile. Although the poles calculated from components B and C lie close, they are statistically different and probably reside in different magnetic carriers (magnetite and pyrrhotite), what calls for different, discrete pulses of remagnetization along the basin.

Despite the strong remagnetization experienced by these rocks an older direction may have been preserved locally, as suggested by the high-temperature component of site 20 (Fig. 4d). The site mean direction calculated out of five samples gives a virtual geomagnetic pole at 80.0°S, 30.6°E ($\alpha_{95} = 4.2°$, $k = 475$) (Table 2). Although based on quite a few samples, it is worth mentioning its similarity to the pole obtained from the La Tinta Group (80°S, 301°E), in Argentina. An age of ~720 Ma is assigned to these sediments (Valencio et al., 1980) which is comparable to the sedimentation age of the Bambuí carbonates (see above).

6.2. Ages of the poles B and C

The approximate time of remagnetizations recorded by components B and C can be compared to the Pb–Pb isochron ages. Samples collected from one outcrop (BA98-15) showed a good spread of Pb isotopic compositions yielding an isochron age of 514 ± 33 Ma. Other carbonates from deformed as well as undeformed sites did not yield enough spread to generate an isochron. However, if considered together with samples from site BA98-15, they yield ages varying from 522 ± 25 Ma to 486 ± 28 Ma, which are undistinguishable within analytical error. These ages are similar to some of the ages obtained in the southern part of the São Francisco basin (Babinski et al., 1999; D'Agrella-Filho et al., 2000). These results suggest that the Pb isotopic system was simultaneously reset at different areas across the basin, probably due to a large-scale fluid percolation event (see below). Therefore, the geochronological data place the remagnetization events at Cambrian times.

The age of each remagnetization event can be further refined by the comparison of the Salitre and Bambuí poles with reference Cambrian poles of Gondwana after rotation of South America to Africa

Table 4
Selected paleomagnetic poles for Gondwana between 550 and 500 Ma

(Number, continent) Pole	Plat. (°N)	Plong. (°E)	α_{95} (°)	Age (Ma)	Reference
(1, AF) Sinyai Dolerite	−28.4	319.1	5.0	547	Meert and Van der Voo (1996)
(2, AF) Mirbat SS	−31.9	333.9	7.2	550	Kempf et al. (2000)
(3, AU) Australian mean pole #1	−9.4	334.1	8.4	535	See note
(4, SA) Bambuí C	31.9	339.1	3.8	525	D'Agrella-Filho et al. (2000)
(5, SA) Salitre C	34.8	338.4	4.0	525	This study
(6, SA) São Francisco mean C	33.6	338.7	2.8	525	This study
(7, AF) Ntonya Ring Structure	27.7	344.9	1.8	522	Briden et al. (1993)
(8, MA) Madagascar virgation zone	12.2	351.1	14.2	521	Meert et al. (2003)
(9, AU) Australian mean pole #2	25.0	349.7	5.7	520	See note
(10, SA) Sierra de las Animas	33.2	359.8	18.1	520	Sanchez-Bettucci and Rapalini (2002)
(11, AS) Bambuí B	26.9	358.9	2.5	515	D'Agrella-Filho et al. (2000)
(12, SA) Salitre B	43.6	002.7	10.2	515	This study
(13, AU) Australian mean pole #3	14.2	358.6	4.0	510	See note
(14, AS) Juiz de Fora complex	12.1	357.5	10.3	510	Raposo et al. (2003)
(15, MA) Carion Granite	12.7	359.7	11.0	508	Meert et al. (2001)
(16, AN) Sor Rondane	10.6	008.3	4.5	510	Zijderveld (1968)
(17, SA) Piquete Fm.	23.0	022.0	10.2	510	D'Agrella-Filho et al. (1986)

Australian mean poles were calculated from #1, Lower Arumbera SS, Brachina Fm., Upper Arumbera SS, Todd River Dolomite; #2, Bunyeroo Fm., Hawker Group A, Hawker Group B; #3, Aroona-Wirealpa A, Aroona-Wirealpe B, Tempe Fm., Hudson Fm., Lake Frome A, Lake Frome B, Giles Creek Dolomite Lower, Giles Creek Dolomite Upper, Illara SS, Deception Fm. (poles recalculated by Meert et al., 2001). Poles are rotated to Africa (AF) according to the following Euler rotation poles (Lowver and Scotese, 1987): AU (25.1°N, 110.1°E, −56.7°), SA (45.5°N, −32.2°E, +58.2°), MA (−3.41°N, −81.7°E, +19.7°), AN (−7.78°N, −31.42°, +58.0°). AU (Australia), SA (South America), MA (Madagascar), AN (Antarctica).

(Table 4). For that, we assume the Gondwana supercontinent had already assembled by that time (e.g. Meert and Van der Voo, 1996; Meert et al., 2001; Meert, 2003). The poles of Fig. 10 (numbers refer to Table 4) follow a sinuous APW path, which swings around the ∼547 Ma Sinyai Dolerite pole (1), the ∼535 Ma Australian mean pole (3), the ∼520 Ma Australian mean pole (9), and the ∼510 Ma Sor Rondane pole (16). The mean C pole (6) plots away from the 550 to 530 Ma poles (Fig. 10). It falls right to the west of the 520 Ma APW path segment, where the mean ∼520 Ma Australian pole (9) and the Ntonya Ring pole (7) are found. Thus, an age slightly older than 520 Ma is suggested for component C magnetization of Salitre and Bambuí. The poles B (SaB and BaB) plot to the east of the ∼520 Ma Australian (9) and Ntonya Ring (7) poles. But they roughly coincide with the ∼520 Ma Sierra de las Animas pole recently determined for the Rio de la Plata craton (Sanchez-Bettucci and Rapalini, 2002). Nonetheless, they are significantly different from the ∼510 Ma Sor Rondane pole, and from the 508 ± 11 Ma Carion Granite pole (Meert et al., 2001). They are also distinct from the ∼510 Ma Juiz de Fora and Piquete poles, obtained for high-grade metamorphic rocks of the Ribeira belt, southeastern Brazil (D'Agrella-Filho et al., 1986; Raposo et al., 2003). Hence, the position of pole B is compatible with an age of ∼520 Ma, being slightly younger than the pole C.

The poles of São Francisco craton carbonate rocks confirm the z-shaped pattern for the Cambrian APW path of Gondwana as recently proposed by Meert (2003). If the apparent polar movement is attributed only to the translation of the Gondwana supercontinent, this path indicates a fast movement for the whole supercontinent between 550 and 510 Ma. Considering the tying points indicated above (poles 1, 4, 9 and 16; Fig. 10), the plate velocity would be of 22 cm per year (∼2.0° Ma^{-1}) between 547 and 535 Ma; 28 cm per year (∼2.5° Ma^{-1}) between 535 and 520 Ma, and 25 cm per year (∼2.3° Ma^{-1}) between 520 and 510 Ma.

Cordani et al. (2000) summarized the events that gave birth to the South American platform. Ages of Neoproterozoic–Cambrian collisions range from 900 Ma up to 500 Ma. Although this age interval has

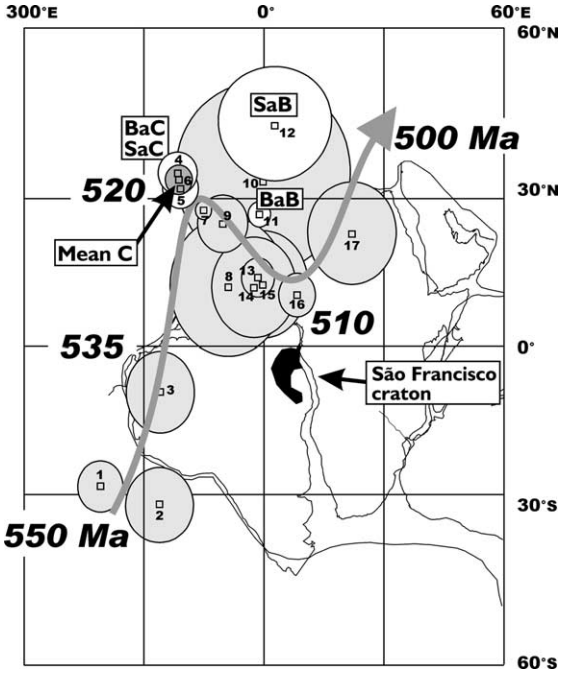


Fig. 10. Paleomagnetic poles of São Francisco carbonates (SaB, SaC, BaB, BaC, and mean C pole) in an African reference frame and APW path for Gondwana between 550 and 500 Ma. For numbers, poles and Euler rotations to Africa refer to Table 4.

been classically referred to a unique, anomalously long orogenic event, the so-called Brasiliano Orogeny (e.g. Brito Neves and Cordani, 1991), recent contributions recognize discrete orogenic phases along this time span. The Brasiliano Orogeny is thus better described as a protracted orogenic episode, similarly to the recent descriptions of the East African Orogen (Stern, 2002; Meert, 2003). Three main peaks of deformation, metamorphism and granite emplacement are reported in the South American Platform at 640–630, 580–570 and 530–500 Ma (e.g. Brito Neves et al., 1999). This later, Cambrian orogenic pulse has been characterized in more detail at southeastern Brazil (Rio Doce orogeny: Campos Neto and Figueiredo, 1995; Buzios orogeny: Schmitt et al., 1999), and western Argentina (Pampean orogeny: Rapela et al., 1998). It results from terrene accretions across the western Gondwana during the final stages of supercontinent assembly. Remagnetizations inside the São Francisco craton are coeval to, or slightly post-date, this late-Brasiliano (Rio Doce, Buzios, Pampean?) orogenic phase.

6.3. Remagnetization and mineralization

Large-scale remagnetization events within foreland basins are usually attributed to the influence of orogenesis at the border of cratonic areas (Oliver, 1986). In such model, deformation at the orogenic front would expel reactive fluids towards the basin, promoting widespread CRM in the sediments. Numerical modeling, however, has shown that the fluid migration induced by tectonic deformation is only effective near the orogenic front, being too slow to drive fluids to the center of the basin (Ge and Garven, 1992). Alternatively, if sufficient hydraulic head can be provided by the mountain chain, gravity-driven brines would flow at the high rates necessary to promote large-scale remagnetizations, as well as massive Pb–Zn mineralizations and hydrocarbon migration (Bethke and Marshak, 1990; Garven, 1995).

The gravity-induced flow model would favorably account for the extent and timing of the Cambrian remagnetizations across the São Francisco basin. By that time, three mountain belts surrounded the São Francisco craton: the Brasilia belt to the west, the Araçuai belt to the east, and the Riacho do Pontal-Sergipano belt to the north (Fig. 1). In all of them, peak metamorphic conditions were attained before 550 Ma ago (Brito Neves et al., 1999). Therefore, at the time of remagnetizations the relief demanded for regional fluid flow had already been available. However, in addition to these basinal brines, some amount of fluid must have migrated through the basement. Sulphide mineralizations (galena and pyrite) hosted by the Neoproterozoic Salitre carbonates present radiogenic Pb signatures which yield an apparent Pb–Pb isochron age of 2138 ± 49 Ma (Toulkeridis et al., 1999; Misi, 1999), indicating that the mineralizing fluid has an old source. Anomalous Archean to Paleoproterozoic Pb signatures are also observed in sulphide mineralizations (Iyer et al., 1992) and the host carbonates of the southern part of the basin (Babinski et al., 1999). The similarity among Pb signatures in both carbonates and sulphides indicates a common source for the fluids responsible for the Pb–Zn mineralization and remagnetization. A regression line drawn with Pb isotopic data of non-radiogenic and radiogenic (old) crustal Pb for these carbonates intercepts the Stacey and Kramers (1975) Pb evolution curve at 520 and 2100 Ma. We assume that these ages track the Pb evolution that took

place in the Archean to Paleoproterozoic basement, which was strongly affected by the 2.1 Ga Transamazonian orogeny, until its introduction into the carbonate rocks at 520 Ma ago (Babinski et al., 1999; D'Agrella-Filho et al., 2000). Thus, mineralization and remagnetization events must overlap in time. Indeed, fluid inclusions and sulfur isotope from some of the sulphide and willemitic deposits indicate that mineralization is due to meteoric fluids mixed with hot, saline fluids that migrated through basement faults (Monteiro et al., 1999; Monteiro, 2002). A similar mechanism could likely account for the remagnetization process.

7. Conclusion

Neoproterozoic carbonates collected from different areas of the São Francisco and Irecê basin (São Francisco craton) showed similar Pb–Pb isochron ages and paleomagnetic poles. These rocks have recorded a widespread remagnetization, with characteristic directions carried by magnetite and pyrrhotite. The Pb–Pb isochrons and the comparison of the poles with reference poles of Gondwana place the remagnetization and the reset of the isotopic system at Cambrian times. It is suggested that saline fluids that migrated across the basin at around 520 Ma, at the end of the late-Brasiliano (Rio Doce, Buzios, Pampean?) orogeny, may have triggered such a continental-scale remagnetization.

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